Electrodynamics & Optics

1

Topic:

1. Static eletricity, electric forces
2. Electric currents
3. Magnetic forces
4. Time dependent electromagnetic fields

   Milestone: Maxwell's eqyations
5. Electromagnetic Wave propagation
6. Optics

(Math: Integration & differentiation of vector fields)

Course description (Schedule): projector

Ancient Greeks recorded the first experiments with electricity (Thales ~600 BC)

Amber (a natural "rock" like material, fossilized pine-tree resin) called

(1) Rubbing some materials with some other materials: magic forces appear

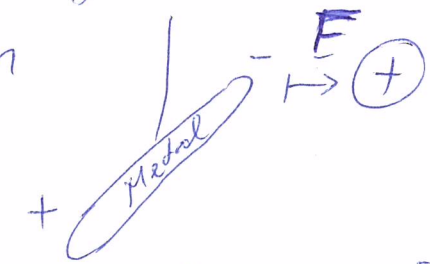
(2) Forces can be attractive or repulsive.)
Two kinds of triboelectric forces Amber-Amber repulsive
Amber-fur attractive

(3) Benjamin Franklin (18. th century): let's denote it by + & −

(4) We can not create electrical charge!
Rubbing, movement etc. devides + & − charges.
~~Neutral~~ Neutral object: same quant. of + & −
(Conservation of electric charge!!)

(5) Two (three) basic families of materials:
conductors, insulators (+semi + super + ...)

(6) Electric forces by ~~induction~~ redistributing (division) induction
and polarization



(7) 1909: Electric charge is quantized,
integer multiple of elementary charg e
$Q = \pm N \cdot e$ (Robert Millikan)

(8) Superposition principle:
Vectorial sum of forces gives macroscopic force

Origin of triboelectric effects:

- who can explain?

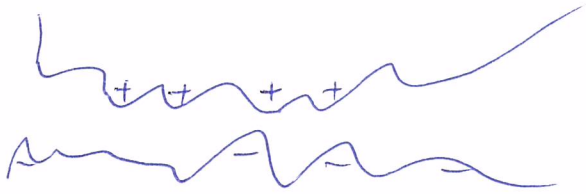
(1) Two materials in contact:

(2) Difference in "electronegativity" of materials

(3) Local exchange of charge (typically e^-).

(4) Insulators: charge remains there!

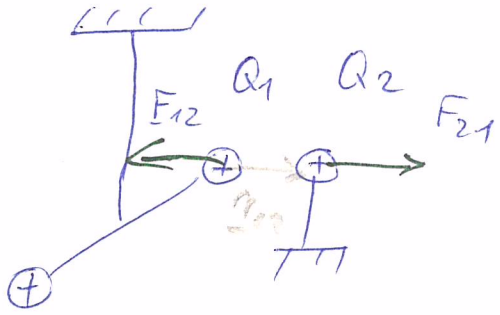
(5) Rubbing is only needed to increase surface!



(6) Similar effect can be observed in conductors, and liquids (Volta - column, battery)

Coulomb's experiment

(↑ Charles Augustine de ~, 1785)



$$|F_{21}| = |F_{12}| \sim \frac{Q_1 Q_2}{r_{12}^2}$$

unit vector $\underline{e}_{12}$

$$\underline{F}_{21} = -\underline{F}_{12} = K_e \frac{Q_1 Q_2}{r_{12}^2} \cdot \frac{\underline{r}_{12}}{r_{12}}$$

force – counter force

~~Important observations:~~ We have 2 options to define the unit of charge and ~~Re~~ Coulomb's constant K_e

(1)

Choose K arbitrarily (CGS system of units)

$K_e = 1$

$$F = \frac{Q_1 Q_2}{r^2}$$

$$[F] = [m \cdot g] = g \cdot \frac{cm}{s^2}$$

$$Q = \sqrt{F \cdot r^2} \quad \& \quad [Q] = \left[g \frac{cm}{s^2} \cdot cm^2 \right] =$$

$$= g^{1/2} cm^{3/2} s^{-1} = 1 Fr$$

1 Franklin

II. Define the unit of charge anyway

→ Measure K_e

in ~~1919~~ 2019 SI unit system defined

1C as the charge of 6.24×10^{18} electrons or protons

↳ i.e. $e = 1,60218 \times 10^{-19} C$

from Coulomb's law we can measure:

$$K_e = 8,9876 \times 10^9 \frac{Nm^2}{C^2}$$

$K_e := \frac{1}{4\pi\varepsilon_0}$

$\varepsilon_0 = 8,8542 \cdot 10^{-12} \frac{C^2}{Nm^2}$

Who knows the definition before 2019??

Important observations:

- Experimental accuracy of Coulomb's original measurement shows $r^{-2 \pm \varepsilon}$ $\varepsilon \sim$ few % [today 10^{-16}]

Ch 1 - Can you estimate the accuracy of the exp. by considering the finite size of the spheres?

Ch 2 proof!

- How we know that $\varepsilon = 0$?

If and only if $\varepsilon = 0$ $\left(F \sim \frac{1}{r^2}\right)$ $\Longleftrightarrow$ $E = 0$ in a conductor

Ch 3: What is inside the charge detecting black box?

- Very similar to gravity: $F_g = \gamma \frac{M_1 M_2}{r^2} \cdot e_{12}$

? differences: - gravity only attractive (not repulsive)

- ~~Magnitude~~ Strength Hatom ⊕ ⊖ $F_e = 8 \cdot 10^{-8}$

- ~~Why~~ Do El. forces influence stellar dynamics? $F_g = 8 \cdot 10^{-47}$ planetary stat

Ch 3 How would the movement of planet Earth change if we removed "only" 1% of its electric charge and transported to the ~~Moon~~ Sun?
Why is this impossible?

Ch 4

Story: Charles Augustine de Coulomb 1785

before him: Priestley (1967 theory)
Cavendish (not published)
Robison (incorrect results, bad accuracy)

Analysis Model: Particle in a field

$$\underline{F}_e = \underbrace{\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\cdot\underline{e}_{12}}_{\underline{E}=\frac{\underline{F}_e}{q}}\cdot q$$

↑ small, positive test charge

Why small?
To avoid division, (induction) of charge(s) in Q
(Idea of M. Faraday)

Electric field vector

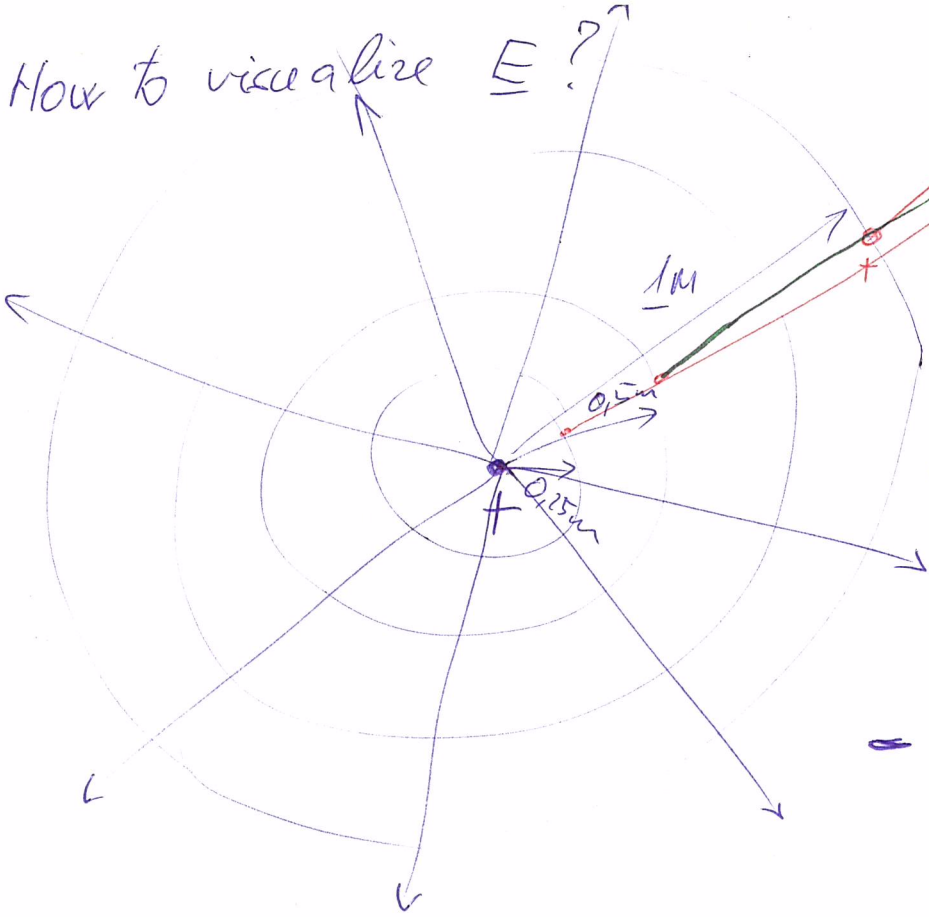
$$[E] = \frac{N}{C}\left(=\frac{V}{m}\text{ later on}\right)$$

~~Electro~~ Static electricity can be well described by remote interaction of charged objects. Serious complications arise, when it comes to EM induction, waves etc.

El. field due to multiple point charges:

$$\underline{E} = \frac{1}{4\pi\varepsilon_0}\cdot\sum_i \frac{q_i}{r_i^2}\underline{e}_i$$

How to visualize $\underline{E}$?



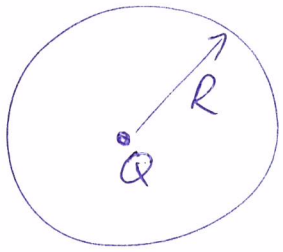
Instead:
electric field lines!

- field lines are $\parallel$ (tangent) to the local $\underline{E}$.
- density of field lines $\sim |E|$

We are lucky again with the $\frac{1}{r^2}$ Coulomb force

why: Let's calculate the ~~density~~ No of field lines for a point charge!

density of field lines $\sim E$



$No \sim \frac{Q\,q}{R^2}$. Surface $\sim R^2$

↳ Number of field lines is constant!

field lines start at ⊕ charges, end at ⊖.

No field line ending is observable in the vacuum!